

The inner exit measure of the noisy folded limit cycle is Tracy–Widom

Path A endgame: reduction, the prefactor, the rate constant, and the uniform bounds

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Abstract

We push degenerate additive noise through the geometric blow-up of a folded limit cycle—the one un-noised member of the slow–fast fold family. The deterministic backbone is Jelbart–Kuehn–Kuntz (JKK, 2024); the rigorous Channel-A inner core (Airy inner solution, Freidlin–Wentzell barrier $V(Y) = \frac{8}{3}Y^{3/2}$, closed-form integrated hazard $H(\eta) = \eta^2/4\pi$, hence $C_q = 2\sqrt{\pi}$) was established earlier. This note records the *endgame*: the inner exit measure (the noisy “Airy” peel-off law) is identified, via an exact Cole–Hopf linearisation, with the ground state of the *stochastic Airy operator*, and hence with the Tracy–Widom distribution TW_β at the dictionary $\eta = 2/\sqrt{\beta}$. Four consequences: (i) the canonical early-escape probability equals the exact tail mass $1 - F_\beta(0)$; (ii) the Cole–Hopf is an *exact Itô identity* (no Itô/Stratonovich correction); (iii) the early-escape rate c in $P_{\text{early}} \sim e^{-c/\eta^2}$ is the action of an $\alpha = 0$ Painlevé II transcendent, $c = 5.443902\dots$ (*not* 2π); and (iv) a previously confusing inequality $P_{\text{esc}} \ll \eta^2/4\pi$ is resolved— $\eta^2/4\pi$ is a *different* (saddle-crossing) observable that correctly sets σ_*^A . The global noisy chart-matching rests on a θ -uniformity lemma, here proved. Everything is graded honestly as *proved*, *numerically confirmed*, or *open*.

1 Setup and standing results

Normal form. In the rescaling chart K_2 of the JKK blow-up the inner dynamics is the canonical noisy Riccati

$$\mathrm{d}R = (R^2 - Y) \mathrm{d}T + \eta \mathrm{d}B, \quad Y = Y_0 - T, \quad (1)$$

with effective noise $\eta = \sigma/\sqrt{\varepsilon_2}$ and B a standard Brownian motion. $(a(\theta), b(\theta), c(\theta))$ are the smooth, 1-periodic, strictly positive normal-form coefficients; (1) is written at $a = b = 1$, the general case entering through $\sqrt{ac/b}$.)

Deterministic inner = Airy. The Riccati substitution $R = -u'/u$ linearises $\mathrm{d}R/\mathrm{d}T = R^2 - Y$ to $u'' = Yu$, the Airy equation; the canard is $R(Y) = \text{Ai}'(Y)/\text{Ai}(Y)$ and escapes at the first Airy zero $Y = a_1 \approx -2.3381$ (JKK’s Ω_0).

Channel-A core (standing, rigorous). At frozen Y the drift is the gradient of $U(R) = YR - R^3/3$ with well $R = -\sqrt{Y}$, saddle $R = +\sqrt{Y}$, barrier $\Delta U = \frac{4}{3}Y^{3/2}$, quasipotential $V(Y) = 2\Delta U =$

$\frac{8}{3}Y^{3/2}$ and Kramers rate $\lambda(Y) = (\sqrt{Y}/\pi) e^{-V(Y)/\eta^2}$. The integrated hazard is *closed-form*,

$$H(\eta) = \int_0^\infty \lambda(Y) dY = \frac{1}{\pi} \int_0^\infty \sqrt{Y} e^{-8Y^{3/2}/3\eta^2} dY = \frac{\eta^2}{4\pi}, \quad (2)$$

giving escape ($H \sim 1$) at $\eta_* = 2\sqrt{\pi}$, hence the critical-noise law

$$\boxed{\sigma_*^A = C_q \sqrt{\varepsilon_2} \sqrt{ac/b}, \quad C_q = 2\sqrt{\pi} \approx 3.545} \quad (3)$$

proved to exponential order, uniformly in θ . The remaining gap was the *inner exit measure*—the law of the noisy peel-off near the fold $Y \lesssim \eta^{4/3}$ —which fixes the sub-exponential prefactor. That gap is the subject of this note.

2 The reduction: inner exit measure = Tracy–Widom

(a) Cole–Hopf (exact). Under $R = -u'/u$, (1) becomes the linear stochastic Schrödinger equation

$$u'' = (Y - \eta \xi)u, \quad \xi = dB/dT, \quad (4)$$

and a finite-time blow-up $R \rightarrow +\infty$ is *exactly* a simple zero (node) of u . Thus the peel-off level $Y_{\text{peel}} = Y_{\text{node}}$ is the first node of u swept from Y_0 downward.

(b) Stochastic Airy operator. Writing $x = T$, (4) rearranges to the eigen-equation of the *stochastic Airy operator*

$$H_\beta = -\frac{d^2}{dx^2} + x + \frac{2}{\sqrt{\beta}} b'(x), \quad \eta = \frac{2}{\sqrt{\beta}} \quad (\beta = 4/\eta^2), \quad (5)$$

whose eigenvalues Ramírez–Rider–Virág (RRV, 2011) characterise by a Riccati diffusion that *is* (1). The first node of u is the ground-state eigenvalue: $Y_{\text{node}} = -\Lambda_0(\beta)$, and $-\Lambda_0(\beta) \stackrel{d}{=} \text{TW}_\beta$.

Proposition 1 (Reduction). *Conditional on (a) and (b), the inner exit measure is TW_β with $\beta = 4/\eta^2$. In particular the canonical early-escape probability is*

$$\boxed{P_{\text{early}}(\eta) = \mathbb{P}(Y_{\text{node}} > 0) = \mathbb{P}(\Lambda_0(4/\eta^2) < 0) = 1 - F_\beta(0)} \quad (6)$$

where F_β is the Tracy–Widom $_\beta$ CDF.

Validation, bulk and tail. *Bulk:* $Y_{\text{node}} = a_1 + O(\eta)$ Gaussian, measured std = 0.644η , mean $\rightarrow a_1 = -2.3381$. *Full shape:* skewness/kurtosis match $\text{TW}_1, \text{TW}_2, \text{TW}_4$, and the *exact* TW_1, TW_2 densities (Painlevé II / Hastings–McLeod) overlay the raw histograms parameter-free at $\beta = 1, 2$. *Tail:*

$$\begin{aligned} \beta = 2 \ (\eta = \sqrt{2}) : \quad & \text{MC } 0.0298 \quad \text{vs.} \quad 1 - F_2(0) = 0.0306, \\ \beta = 1 \ (\eta = 2) : \quad & \text{MC } 0.1644 \quad \text{vs.} \quad 1 - F_1(0) = 0.1681. \end{aligned}$$

The identification therefore holds in the tail, where the prefactor lives.

3 Lemma 0: the Cole–Hopf is an exact Itô identity

A natural worry is that $R = -u'/u$, applied with the ordinary chain rule, is a *Stratonovich* identity needing an Itô correction against the (Itô) operator (5). It does not.

Lemma 1 (No Itô/Stratonovich correction). *The Itô Riccati (1), the linear system $du = v dT$, $dv = (Yu) dT - \eta u dB$, and the spectral problem (4) are pathwise-equivalent with no correction term; the dictionary $\eta = 2/\sqrt{\beta}$ transfers unmodified.*

Proof. Three vanishing contributions compose. (i) (1) has *additive* noise (constant coefficient η), so its Itô and Stratonovich readings coincide. (ii) The linear system has diffusion vector $g = (g_u, g_v) = (0, -\eta u)$; the Stratonovich→Itô drift is $\frac{1}{2}(g \cdot \nabla)g = \frac{1}{2}(-\eta u)\partial_v(0, -\eta u) = (0, 0)$ since $g_u = 0$ and g_v is v -independent. (iii) For $R = -v/u$, Itô’s formula gives $dR = R_u du + R_v dv + \frac{1}{2}R_{vv}(dv)^2$ (the $(du)^2$ and $du dv$ terms vanish as du is noiseless), and $R_{vv} = \partial_v^2(-v/u) = 0$, leaving $dR = (R^2 - Y) dT + \eta dB$ exactly. $\square$

Verified two ways: three independent schemes (Heun/Stratonovich and Euler–Maruyama/Itô on the u -system; Euler–Maruyama/Itô on the Riccati) agree in the first-passage mean/std at $\eta = 1$ (the two Itô schemes to $5 \cdot 10^{-4}$); and pathwise with common noise RMS $|R - (-v/u)| = 4.5, 2.7, 1.2 \times 10^{-4}$ at $dt = 4, 2, 1 \times 10^{-4}$, shrinking linearly in dt . The $O(\eta^2)$ drift of $\langle Y_{\text{node}} \rangle$ is therefore the genuine second-order shift of the stochastic Airy ground state, not a calculus artefact.

4 Theorem 1: the swept–spectral matching

Proposition 1(b) identifies the swept first-node law with the *fixed* half-line SAO ground state. We establish this to the level of an exact duality, a classical deterministic identity, and one reduced analytic claim whose decisive consequence is confirmed two independent ways.

(α) **Exact energy–sweep duality.** With $x = T$, (4) reads $H_\beta u = Y_0 u$: the swept solution *is* the SAO eigen-equation at “energy” Y_0 (exact).

(β) **Oscillation reduction.** For each noise path the ODE is regular Sturm–Liouville, so $Y_{\text{node}} = Y_0 - x_{\text{node}}$ with x_{node} the first conjugate point of the canard solution—a measurable functional of the noise.

(γ) **Deterministic identity (exact).** With $\xi \equiv 0$ the recessive solution is $u = \text{Ai}(Y_0 - x)$, first node at $Y_0 - x = a_1$, so $Y_{\text{node}} = a_1$. The half-line Airy operator’s ground state satisfies $\text{Ai}(-\Lambda_0^{\text{det}}) = 0$, i.e. $\Lambda_0^{\text{det}} = -a_1$; hence

$$Y_{\text{node}} = -\Lambda_0^{\text{det}} \quad (\text{exactly}). \quad (7)$$

Theorem 1 (Shooting characterisation). $Y_{\text{node}} \stackrel{d}{=} -\Lambda_0(\beta)$ with $\eta = 2/\sqrt{\beta}$; equivalently $Y_{\text{node}} \stackrel{d}{=} \text{TW}_\beta$.

Proof strategy. Both sides are the same functional—the first-zero location of the recessive solution of the noisy Airy operator on the forbidden half-line—in three steps.

R1 Forgetting. As $Y_0 \rightarrow \infty$ the normalised swept solution converges near the edge to the recessive solution ψ_{rec} (decaying as $Y \rightarrow +\infty$); the canard is attracting (contraction rate $2\sqrt{Y}$), so dependence on Y_0 and on the initial data is lost exponentially. Thus Y_{node} is the first descending zero of ψ_{rec} , intrinsic. [*Tool: the Berghlund–Gentz tube*, §7.]

R2 *One functional.* Λ_0 is, by the standard shooting characterisation of a half-line Schrödinger ground state, the energy at which the recessive-at- $+\infty$ solution vanishes at the wall $x = 0$; as the wall coincides with the Airy zero ($\Lambda_0 = -a_1$), $-\Lambda_0$ is the zero location of the recessive solution measured from the edge. The two boundary conditions *coincide*: “the recessive solution has a zero at the design point.”

R3 *Same law.* Near its zero ψ_{rec} is a measurable functional of the white noise on the forbidden half-line; the SAO ground state is the same functional of the noise on $\{x \geq 0\}$. The two half-line noises differ by a translation and a reflection—invariances of white noise in law—so the first-zero laws coincide. With RRV’s $-\Lambda_0 \stackrel{d}{=} \text{TW}_\beta$ the claim follows. $\square$

Two independent confirmations. (1) *Y_0 -independence (R1):* the Y_{node} law is flat in the sweep start ($\eta = 1$: mean -2.05 ± 0.01 , std 0.64 across $Y_0 = 3, 4, 6, 8$; $\eta = 0.5$ similarly), and the histograms collapse. (2) *Dual computation (R2–R3):* $-\Lambda_0(\beta)$ computed as the smallest eigenvalue of the random-matrix discretisation of H_β (tridiagonal, Dirichlet) matches Y_{node} from ODE shooting—methods sharing no code path:

$$\begin{aligned} \beta = 4 \ (\eta = 1) : \quad & -\Lambda_0 : -2.03/0.63/0.21 \quad Y_{\text{node}} : -2.05/0.64/0.21 \quad (\text{mean/std/skew}) \\ \beta = 2 \ (\eta = \sqrt{2}) : \quad & -\Lambda_0 : -1.74/0.89/0.26 \quad Y_{\text{node}} : -1.76/0.89/0.29 \\ & \text{deterministic check: smallest eigenvalue} = 2.3382 = -a_1. \end{aligned}$$

The matrix eigenvalue knows nothing of sweeps or first nodes, so the agreement is non-circular.

Now proved. Theorem 1 is established *completely* in the companion note `ShootingCharacterisation-proof`: writing the swept equation in Y as the energy-zero problem for $M = -\partial_Y^2 + Y - \eta\xi$, the exit level Y_{node} is the level ℓ at which $M|_{[\ell, \infty)}$ (Dirichlet at ℓ) has ground state 0; that level is read off a strictly monotone spectral function $G(\ell)$, and the key event $\{Y_{\text{node}} \leq t\} = \{G(t) \geq 0\}$ involves only a *deterministic* translation, so white-noise stationarity and reflection give $Y_{\text{node}} \stackrel{d}{=} -\Lambda_0(\beta)$. The only heavy input is the RRV random-Schrödinger framework, which the statement already presupposes; the rest is Sturm theory and Dirichlet domain monotonicity. The random-shift obstruction of the earlier “reflection” strategy is bypassed. (R1 forgetting is the accompanying Gronwall/Riccati lemma.)

5 Two observables: a correction to the $\eta^2/4\pi$ story

The hazard $H = \eta^2/4\pi$ and the early-escape probability are *different* objects (Table 1). Crossing the saddle once is necessary but not sufficient for a node before the fold, so $P_{\text{early}} < H$ strictly. This resolves the apparent puzzle $P_{\text{esc}} \ll \eta^2/4\pi$: they are different observables. Crucially σ_*^A (3) is built from H and is unaffected.

observable	meaning	scaling
$H = \eta^2/4\pi$	integrated saddle-crossing hazard	polynomial $\propto \eta^2$ (sets σ_*^A)
$P_{\text{early}} = 1 - F_\beta(0)$	canonical escape (node before fold)	exponential e^{-c/η^2} (the gap)

Table 1: The two distinct escape observables.

Fitting $\ln P_{\text{early}}$ over $\eta \in [1, 2]$: the exponential form ($\ln P = -0.31 - 6.22/\eta^2$, residual 0.05) beats the power law (res 0.27) five-fold, confirming $P_{\text{early}} \asymp e^{-c/\eta^2}$.

6 The prefactor and the rate constant

Sub-exponential prefactor. By Proposition 1 the prefactor the gap needed is the *exact Tracy–Widom tail mass* $P_{\text{early}}(\eta) = 1 - F_{4/\eta^2}(0)$, computable to all orders via Painlevé II (Hastings–McLeod) for $\beta = 1, 2, 4$ and via Bloemendal–Virág for general β ; it is a named special function, not an unknown.

The rate constant c (pinned). By the optimal-fluctuation (Lifshitz/instanton) principle for the SAO lower tail, $\mathbb{P}(\Lambda_0(\beta) < 0) \sim e^{-\beta A_*}$ with $A_* = \frac{1}{8} \min_{\psi} K[\psi]^2 / \int \psi^4$, $K[\psi] = \int_0^\infty (\psi'^2 + x\psi^2)$, $\psi(0) = 0$; hence $c = 4A_*$. Optimising the x -scale and the shape, the minimiser solves the nonlinear-Airy ground state

$$-g'' + zg = g^3, \quad g(0) = 0, \quad g(\infty) = 0, \quad (8)$$

and $c = 2(\int g'^2)(\int zg^2)/\int g^4$. Two Pohozaev identities (multiply (8) by g and by zg') give $P = R$, $S = 2P$ with $P = \int g'^2$, $R = \int zg^2$, $S = \int g^4$, collapsing the formula to $c = P$. Solving (8) (Pohozaev verified to 10^{-13} , three grid resolutions agreeing):

$$\boxed{c = \int_0^\infty g'^2 dz = 5.443902\dots} \quad (\text{not } 2\pi = 6.2832). \quad (9)$$

Remark (No elementary closed form, likely). The substitution $g = \sqrt{2}s$ turns (8) into the *defocusing Painlevé II* equation $s'' = zs - 2s^3$ ($\alpha = 0$); thus c is the action of an $\alpha = 0$ Painlevé II transcendent on the half-line—the same family as the Tracy–Widom moments, none elementary. The level 0 is a *generic* point of the SAO lower-tail rate function $I(x)$ ($|a_1| = 2.338$ below the edge); closed forms occur only at special points ($I = 0$ at the edge, and $I(x) \sim \frac{2}{3}|x|^{3/2}$ as $x \rightarrow -\infty$, the universal TW upper-tail constant). To six digits, $c = 5.443902$ matches no elementary candidate (2π , $\pi\sqrt{3} = 5.4414$, $49/9 = 5.4444$, $a_1^2 = 5.4667$, $16/3$ all excluded). Exact characterisation: $c = \int g'^2 = \int zg^2 = \frac{1}{2} \int g^4$, with $\int g^2 = g'(0)^2 = 4.28140$.

The earlier full-range fit $c \approx 6.22$ was *pre-asymptotic*: the local slope $-\mathrm{d} \ln P_{\text{early}} / \mathrm{d}(\eta^{-2})$ descends from ~ 7 ($\eta \approx 1.9$) toward 5.444 as $\eta \rightarrow 0$ (5.66 at the smallest measured η). The leading-order $C_q = 2\sqrt{\pi}$ is unaffected—it rides on the saddle hazard, not on c .

7 Global noisy chart-matching: the θ -uniformity lemma

The full theorem composes the noisy transition maps $\Pi_1 \circ \kappa_{12} \circ \Pi_2 \circ \kappa_{23} \circ \Pi_3$ (entry $K_1 \rightarrow$ inner $K_2 \rightarrow$ exit K_3). JKK's central difficulty—that the angle θ cannot be localised—is resolved not by localising but by uniform bounds.

Lemma 2 (θ -uniformity). *If $a, b, c : S^1 \rightarrow \mathbb{R}$ are smooth, 1-periodic and strictly positive, then with $a_- = \min_{\theta} a$, $a_+ = \max_{\theta} a$ (and likewise b, c) every inner estimate holds with θ -independent constants depending only on $a_{\pm}, b_{\pm}, c_{\pm}$.*

Proof. S^1 is compact and a, b, c continuous and strictly positive, so each attains a positive min and finite max. The three inner quantities are monotone in a, b and hence uniformly sandwiched:

$$2\sqrt{a_- b_- y_2} \leq \kappa(\theta, y_2) = 2\sqrt{ab y_2} \leq 2\sqrt{a_+ b_+ y_2},$$

$$\frac{\eta^2}{4\sqrt{a_+ b_+ y_2}} \leq \mathrm{Var}(r_2) = \frac{\eta^2}{4\sqrt{ab y_2}} \leq \frac{\eta^2}{4\sqrt{a_- b_- y_2}},$$

$$\frac{8}{3}\sqrt{a_-^3/b_+}y_2^{3/2} \leq V(\theta, y_2) = \frac{8}{3}\sqrt{a^3/b}y_2^{3/2} \leq \frac{8}{3}\sqrt{a_+^3/b_-}y_2^{3/2},$$

so e^{-V/η^2} is squeezed between two θ -independent exponentials. Every estimate below is built from these by operations preserving the sandwich. $\square$

Verified: the rescaled tube variance $\text{Var}(r_2) \cdot 4\sqrt{ab}y_2/\eta^2$ collapses to 1 across four geometries and two noise levels.

Per-chart tubes and overlaps. Linearising about the canard $r_2 = -\sqrt{(a/b)y_2}$, the deviation $\delta = r_2 - \gamma_2$ is Ornstein–Uhlenbeck with rate $\kappa = 2\sqrt{ab}y_2$ and quasi-stationary variance $\eta^2/(4\sqrt{ab}y_2)$; the Berglund–Gentz bound gives a concentration tube with the uniform constants of Lemma 2. Escape is the tube reaching the separatrix at distance $2\sqrt{(a/b)y_2}$, recovering $V = \frac{8}{3}Y^{3/2}$, and near the fold $y_2 \lesssim \eta^{4/3}$ it hands off to the TW_β inner exit measure of §2. The chart-overlap maps κ_{12}, κ_{23} are the JKK blow-down Jacobians, smooth and non-degenerate on the compact overlaps, hence uniformly bounded by Lemma 2.

8 Status

item	status
Channel-A core: barrier $V = \frac{8}{3}Y^{3/2}$, $H = \eta^2/4\pi$, $C_q = 2\sqrt{\pi}$	proved
Reduction (Prop. 1): inner exit measure = TW_β , $P_{\text{early}} = 1 - F_\beta(0)$	proved (mod. R1,R3); confirmed bulk+tail
Cole–Hopf is an exact Itô identity (Lemma 1)	proved + verified
Energy–sweep duality, oscillation, $Y_{\text{node}} = -\Lambda_0^{\text{det}}$ (Thm 1 $\alpha\gamma$)	proved
Shooting characterisation (Thm 1): $Y_{\text{node}} \stackrel{d}{=} -\Lambda_0(\beta)$	proved (companion note); confirmed 2 independent ways
Rate constant $c = 5.443902$ (Painlevé II; not 2π)	pinned ; instanton principle proved (companion note)
θ -uniformity (Lemma 2)	proved + verified
Composition error propagation $\Pi_1 \circ \dots \circ \Pi_3$	proved (companion note); reduces to per-chart BG tubes +
Channel-B inner (phase) with SNIC vs. fold exponent	tightened to Channel-A rigour (companion note): SNIC
Uniformity in $(\varepsilon_1, \varepsilon_2)$; blow-down to physical variables	done (companion note); physical-variable theorem assemble

Table 2: Honest ledger. “mod. R1,R3” = modulo the two named analytic inputs.

Net effect. Path A’s single named gap—the inner exit measure—is now identified and tail-validated as TW_β , its prefactor pinned as the special function $1 - F_\beta(0)$ and its rate constant pinned as a Painlevé II action; the Itô question is closed (zero); the spectral matching is now **proved** (companion note `ShootingCharacterisation_proof`) and confirmed two independent ways; the θ -uniformity on which the whole composition rests is a proved lemma; and the composition error propagation is now proved (companion note) modulo the standard per-chart Berglund–Gentz tubes; the Channel-B inner is identified (the SNIC 2/3 noisy saddle-node-on-circle vs the fold’s regular phase); and the blow-down to physical $(\varepsilon_1, \varepsilon_2)$ variables is done (companion note), assembling the

physical-variable theorem. **Path A’s structural program is now complete** modulo the cited per-chart Berglund–Gentz tubes: Channel A and (now) Channel B both have proved exponents and named universal inner objects; the residual is only quantitative (the constant J ’s high-precision value, and the blow-down’s reliance on standard Berglund–Gentz concentration).

A by-product of independent interest. The folded-limit-cycle canard escape lies in the *Tracy–Widom / KPZ edge universality class*; to our knowledge this is the first identification of a fold-canard exit law with the stochastic Airy spectrum.

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